

RUSTAQ, Oman

WMO: 412530

Lat: 23.4092N

Lon: 57.4285E

Elev: 1056

StdP: 14.14

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 54.9	(c) 57.1	(d) 28.4	(e) 23.2	(f) 92.2	(g) 32.1	(h) 27.6	(i) 92.3	(j) 17.2	(k) 76.8	(l) 13.4	(m) 71.3	(n) 2.6	(o) 200	(p) 0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 20.7	(c) 112.9	(d) 70.9	(e) 111.2	(f) 71.2	(g) 109.3	(h) 71.4	(i) 83.1	(j) 96.6	(k) 82.2	(l) 95.6	(m) 81.4	(n) 94.7	(o) 7.5	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	162.2	90.2	79.0	156.6	90.6	77.9	150.9	90.5	48.3	97.2	47.3	95.3	46.2	94.9	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.2	(b) 11.8	(c) 9.9	(e) 50.5	(f) 116.7	(g) 3.2	(h) 1.2	(i) 48.2	(j) 117.6	(k) 46.3	(l) 118.3	(m) 44.5	(n) 119.0	(o) 42.2	(p) 119.9		
(a) 14.2	(b) 11.8	(c) 9.9	(e) 45.1	(f) 85.0	(g) 3.3	(h) 0.8	(i) 42.7	(j) 85.6	(k) 40.7	(l) 86.1	(m) 38.9	(n) 86.5	(o) 36.5	(p) 87.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.7	67.7	71.4	77.6	86.2	95.2	98.3	97.8	95.7	92.4	86.3	77.0	70.1
	DBStd	11.62	3.62	4.89	4.85	4.56	3.73	3.73	3.77	3.84	2.92	3.39	3.61	3.31
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	27	17	7	0	0	0	0	0	0	0	0	0	3
	CDD50	12665	550	598	855	1086	1401	1449	1480	1418	1271	1124	810	623
	CDD65	7216	101	185	390	636	936	999	1015	953	821	659	360	161
	CDH74	104252	493	1373	3838	8490	15442	17101	17187	15525	12525	8429	3014	835
	CDH80	68998	55	426	1780	4950	11010	12790	12724	11065	8243	4736	1099	120
Wind	WSAvg	4.3	3.7	4.1	4.5	4.6	4.6	4.7	4.5	4.6	4.3	4.1	3.8	3.6
Precipitation	PrecAvg	5.4	0.5	1.1	0.7	0.7	0.3	0.3	0.4	0.2	0.2	0.5	0.4	0.5
	PrecMax	8.2	2.6	4.5	4.1	2.7	1.5	2.3	1.3	0.6	0.9	3.3	1.5	1.5
	PrecMin	2.2	0.0	0.5	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1
	PrecStd	1.9	0.7	1.0	1.0	0.8	0.4	0.7	0.3	0.2	0.3	0.8	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	92.7	98.2	105.8	114.0	115.7	114.4	112.9	108.9	104.0	93.3	86.0
		MCWB	63.0	64.1	64.3	66.9	69.6	69.8	72.4	72.1	69.8	67.7	65.6	63.7
	2%	DB	81.5	88.5	95.5	102.8	111.0	113.4	112.1	110.5	106.2	101.3	90.5	83.2
		MCWB	63.6	63.6	64.6	66.6	69.2	69.9	73.0	73.3	70.3	67.3	67.0	64.0
	5%	DB	79.1	84.9	92.5	100.4	108.8	111.3	109.8	108.0	104.3	99.1	88.6	80.9
		MCWB	62.9	63.6	64.5	66.0	68.8	70.0	73.5	73.3	71.1	67.0	66.8	64.5
	10%	DB	76.8	81.9	89.5	97.7	106.6	109.2	107.5	105.8	102.3	96.7	86.4	78.8
		MCWB	62.3	63.4	64.1	65.9	68.6	70.6	74.6	74.2	71.9	66.9	66.7	63.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	70.8	73.3	75.4	80.0	83.9	84.6	83.4	81.7	78.6	74.2	70.0
		MCDB	77.4	80.5	83.2	86.6	91.8	96.8	98.1	97.1	93.4	87.4	82.0	76.8
	2%	WB	66.6	69.0	71.6	73.8	77.1	82.2	83.5	82.4	80.4	77.2	72.5	68.6
		MCDB	75.1	78.2	81.7	84.8	90.7	95.3	97.2	95.8	92.2	85.2	81.5	76.1
	5%	WB	65.5	67.7	70.2	72.0	74.7	81.0	82.6	81.5	79.4	75.9	71.1	67.4
		MCDB	73.9	77.0	80.6	84.3	93.0	93.7	96.2	94.8	91.2	84.9	81.1	75.2
	10%	WB	64.3	66.3	68.8	70.3	72.7	79.7	81.8	80.6	78.5	74.5	69.9	66.3
		MCDB	72.8	76.0	80.0	86.1	96.3	93.9	95.6	93.6	90.3	84.6	80.0	74.1
Mean Daily Temperature Range	5% DB	MDBR	17.9	20.0	21.0	21.1	22.2	20.7	18.2	19.4	20.2	20.8	18.3	17.2
		MCDBR	20.5	23.5	24.9	23.9	24.1	23.3	21.0	21.6	22.3	23.1	19.9	19.6
		MCWBR	7.1	7.9	8.0	6.2	6.4	6.5	7.3	6.7	6.9	7.8	6.6	6.5
	5% WB	MCDBR	17.4	18.4	18.5	19.0	20.8	18.3	17.1	18.0	18.1	18.4	17.5	16.0
		MCWBR	7.2	7.3	7.3	7.6	9.0	7.8	7.4	6.6	6.8	7.4	6.4	5.8
Clear-Sky Solar Irradiance	taub	0.463	0.527	0.610	0.678	0.706	0.778	0.842	0.808	0.695	0.563	0.496	0.463	
	taud	2.015	1.847	1.664	1.571	1.528	1.444	1.391	1.426	1.568	1.788	1.948	2.011	
	Ebn at Noon	248	240	227	216	209	193	181	187	205	227	236	242	
	Edh at Noon	50	63	79	88	92	99	104	101	86	66	53	49	
All-Sky Solar Radiation	RadAvg	1466	1718	1991	2197	2395	2369	2205	2154	2027	1844	1532	1387	
	RadStd	74	84	113	132	61	82	82	53	42	56	44	69	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.76	+1.22	N/A	N/A	+0.50	+0.54	N/A	N/A	+173	+176

Nomenclature: See separate page